Deep Learning (UCS761)

Pneumonia Detection from Chest X-Rays Using Deep Learning

Project Report

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Introduction

Pneumonia is a serious respiratory condition that affects the lungs and can lead to severe complications if not diagnosed early. Chest X-ray imaging is one of the most commonly used diagnostic tools for detecting pneumonia. However, manual interpretation of X-ray images is time-consuming and depends heavily on the expertise of radiologists.

With the advancement of deep learning, Convolutional Neural Networks (CNNs) have shown strong performance in medical image analysis tasks. CNN-based models can automatically learn spatial features from images and perform classification with high accuracy. This project focuses on developing deep learning models to classify chest X-ray images into two categories: Normal and Pneumonia.

The dataset used consists of labeled chest X-ray images, which are preprocessed and used to train multiple CNN-based models, including a custom CNN and transfer learning models such as VGG16 and ResNet50. Additionally, more advanced architectures like DINOv2 are explored for performance comparison.

Literature Review

Deep learning has significantly improved performance in medical image classification, particularly in chest X-ray analysis. Various CNN architectures have been proposed and applied to pneumonia detection with promising results.

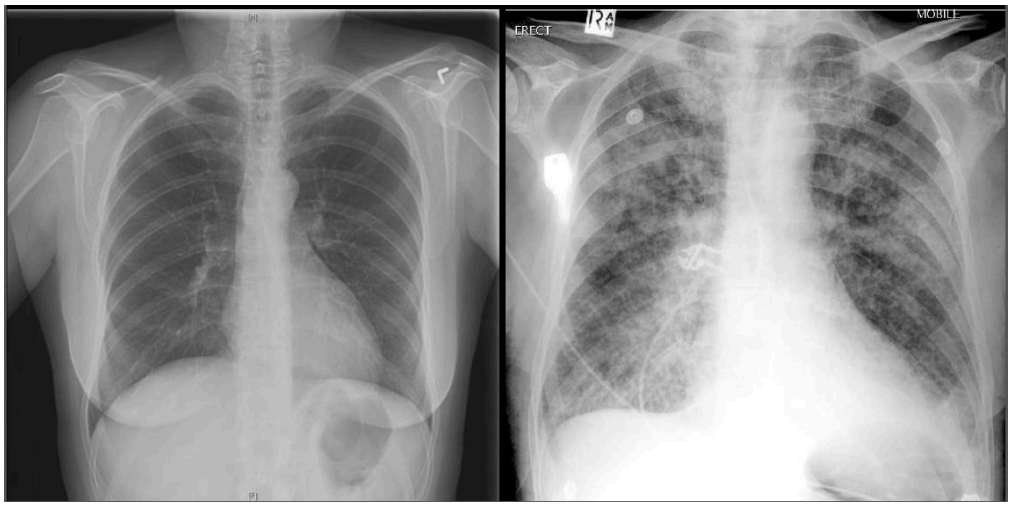

Fig. 1: Normal vs Pneumonia-ridden Chest X-Ray Scans

Kermany *et al.* [1] applied transfer learning techniques to medical imaging tasks and showed that pretrained CNN models can achieve high accuracy even with limited labeled data. This approach has become widely adopted in medical image classification.

He *et al.* [2] proposed ResNet, which uses residual connections to enable training of very deep networks by mitigating the vanishing gradient problem. ResNet has been extensively used in medical imaging due to its strong feature extraction capability.

Simonyan and Zisserman [3] introduced VGGNet, which uses small convolutional filters in deep architectures. Despite its simplicity, VGGNet remains a strong baseline for image classification tasks.

Dosovitskiy *et al.* [4] proposed Vision Transformers (ViT), which apply attention mechanisms to image classification. These models achieve high performance when trained on large datasets.

Oquab *et al.* [5] introduced DINOv2, a self-supervised learning approach that learns robust visual representations without labeled data. The model demonstrates strong generalization capability across various vision tasks and can be effectively fine-tuned for classification problems such as medical image analysis.

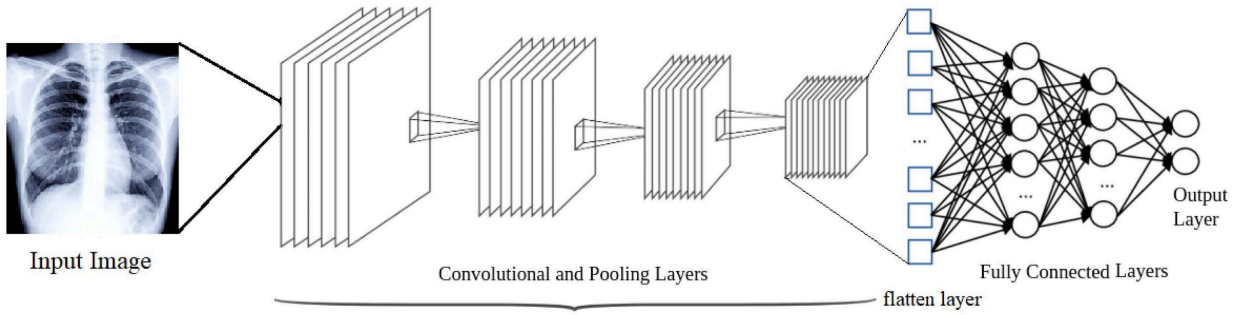

Fig. 2: CNN Architecture Visualization

From the literature, it is clear that:

- Transfer learning significantly improves performance.
- ResNet is highly effective for medical imaging.
- Recent models like DINOv2 provide improved feature representations.

This project builds upon these approaches by comparing multiple architectures on the pneumonia dataset.

Methodology

Models Used

Four different deep learning models were trained to fit the problem of identifying pneumonia-ridden X-ray scans from normal ones. They are as follows:

1. Custom Convolutional Neural Network (CNN)

Model Type

A custom Convolutional Neural Network (CNN) was developed from scratch specifically for the task of binary classification of chest X-ray images into Normal and Pneumonia categories. Unlike pretrained models, this architecture learns all features directly from the dataset, making it a good baseline model to evaluate performance before applying advanced techniques.

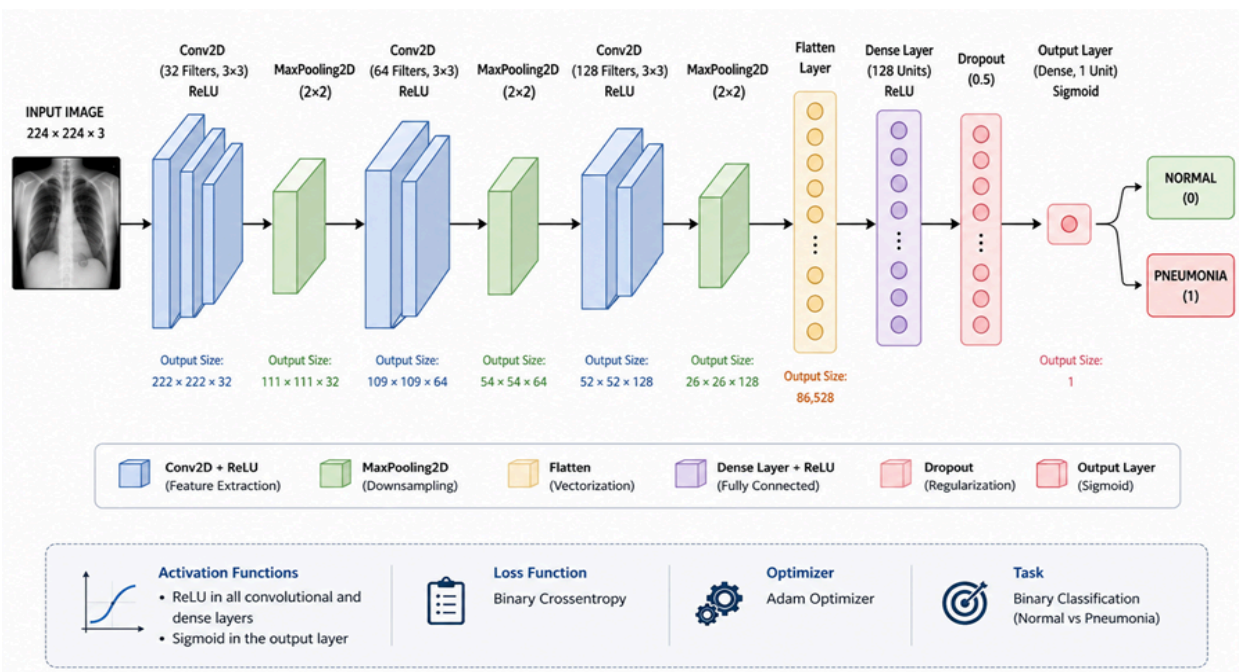

Fig. 3: Custom CNN Architecture

Layers Used

The model architecture consists of multiple convolutional and pooling layers designed to progressively extract features from the images. Initially, a Conv2D layer with 32 filters captures low-level features such as edges and textures, followed by a MaxPooling layer to reduce spatial dimensions. This is followed by additional convolutional layers with 64 and 128 filters, which capture more complex patterns and structures in the X-ray images. Each convolutional layer is followed by a MaxPooling layer to reduce computation and prevent overfitting. The output is then flattened and passed through a fully connected dense layer with 128 neurons. A dropout layer with a rate of 0.5 is used to improve generalization by randomly deactivating neurons. Finally, a single neuron output layer is used for binary classification.

Activation Function

The Rectified Linear Unit (ReLU) activation function is used in all convolutional and dense layers to introduce non-linearity and improve learning capability. In the final output layer, a sigmoid activation function is used to convert the output into a probability value between 0 and 1, suitable for binary classification.

Loss Function

Binary Crossentropy is used as the loss function because it is well-suited for binary classification problems. It measures the difference between predicted probabilities and actual labels, helping the model improve during training.

Optimizer

The Adam optimizer is used for training the model. It is an adaptive learning rate optimization algorithm that combines the advantages of both RMSProp and stochastic gradient descent, making it efficient and widely used in deep learning applications.

2. VGG16 (Transfer Learning)

Model Type

VGG16 is a deep convolutional neural network pretrained on the ImageNet dataset. In this project, it is used with transfer learning, where the pretrained convolutional layers are reused to extract meaningful features from chest X-ray images. This approach significantly reduces training time and improves performance, especially when working with limited datasets.

Layers Used

The convolutional base of VGG16, which consists of multiple convolutional and pooling layers, is used as a fixed feature extractor. All these layers are frozen to retain the learned features from ImageNet. On top of this base, a custom classification head is added. This includes a flatten layer to convert feature maps into a vector, followed by a dense layer with 256 neurons to learn task-specific features. A dropout layer with a rate of 0.5 is applied to reduce overfitting. Finally, a single neuron output layer is used for binary classification.

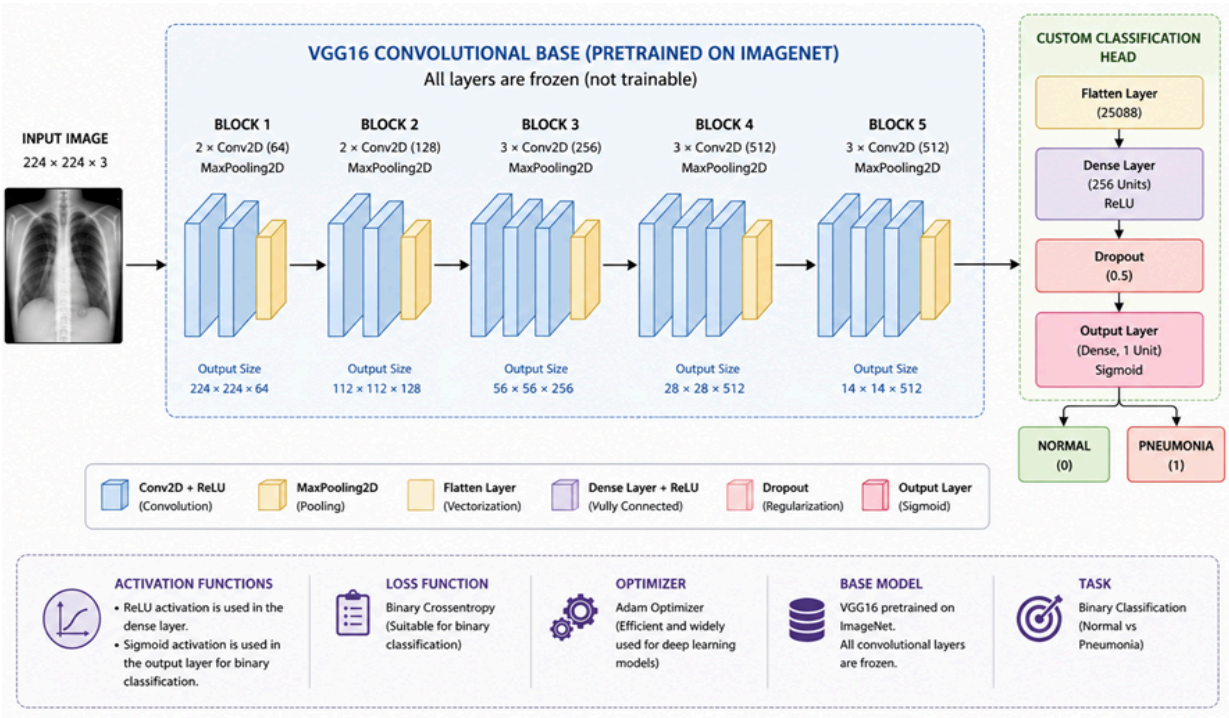

Fig. 4: VGG16 (Transfer Learning) Architecture

Activation Function

ReLU activation is used in the dense layer to introduce non-linearity, while sigmoid activation is used in the output layer to produce a probability value for classification.

Loss Function

Binary Crossentropy is used as the loss function as it is suitable for binary classification problems and helps optimize the model effectively.

Optimizer

The Adam optimizer is used for training due to its efficiency and adaptive learning capabilities. It helps the model converge faster and achieve better performance.

3. ResNet50 (Fine-Tuned Transfer Learning)

Model Type

ResNet50 is a deep convolutional neural network that uses residual connections to allow training of very deep architectures. In this project, it is used with transfer learning and fine-tuning, where some of the deeper layers are allowed to train on the dataset while earlier layers remain frozen.

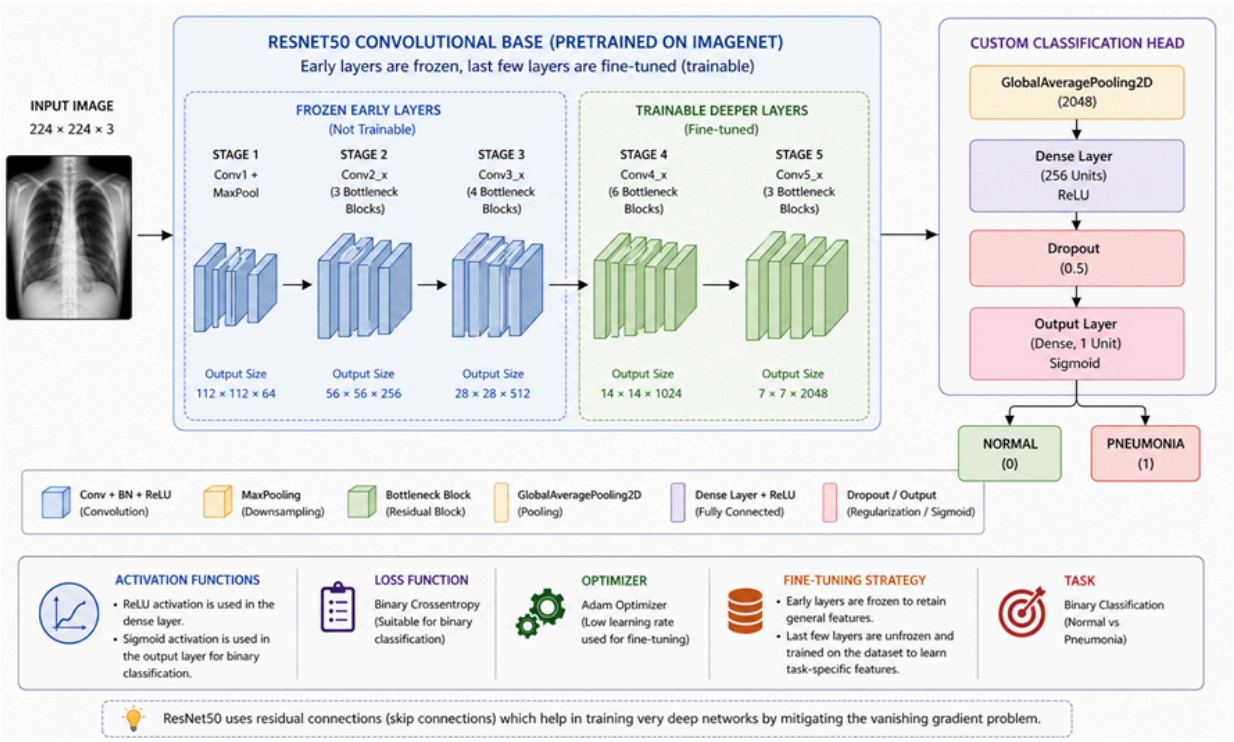

Fig. 5: ResNet50 (Fine-Tuned Transfer Learning) Architecture

Layers Used

The base ResNet50 model pretrained on ImageNet is used. The earlier layers, which capture general features such as edges and textures, are frozen to retain their learned knowledge. The deeper layers are unfrozen to allow adaptation to the pneumonia detection task. A Global Average Pooling layer is used instead of flattening to reduce the number of parameters and

improve generalization. This is followed by a dense layer with 256 neurons, a dropout layer with a rate of 0.5, and a final sigmoid output layer.

Activation Function

ReLU activation is used in the dense layer, while sigmoid activation is used in the output layer for binary classification.

Loss Function

Binary Crossentropy is used as the loss function to measure prediction error in binary classification tasks.

Optimizer

The Adam optimizer with a low learning rate is used to ensure stable fine-tuning. A smaller learning rate helps preserve pretrained weights while allowing gradual adaptation to the dataset.

4. DINOv2-Base (Self-Supervised Transfer Learning)

Model Type

DINOv2-Base is an advanced Vision Transformer (ViT) model pretrained using self-supervised learning. Unlike traditional models such as VGG16 and ResNet50 that rely on labeled data, DINOv2 learns useful visual features without labels during pretraining. In this project, transfer learning is applied by fine-tuning the pretrained DINOv2 model on chest X-ray images for binary classification into Normal and Pneumonia classes.

Layers Used

The DINOv2-Base model uses a Transformer-based architecture consisting of multiple encoder blocks that process image patches instead of pixels. The pretrained backbone is used as a feature extractor. A custom classification head is added on top, which includes a normalization layer, followed by a dense layer with 256 neurons. A dropout layer with a rate of 0.3 is applied to reduce overfitting. Finally, an output layer is used for classification. Training is performed in two stages: initially, only the classification head is trained while keeping the backbone frozen, and then the entire model is fine-tuned.

Activation Function

The GELU (Gaussian Error Linear Unit) activation function is used in the dense layer of the classification head to introduce non-linearity. It is commonly used in Transformer-based

architectures due to its smooth activation behavior. A softmax activation function is applied in the output layer to convert predictions into probability values for classification.

Loss Function

Cross-Entropy Loss is used as the loss function, as it is suitable for classification tasks. Class weighting is applied to handle dataset imbalance, ensuring that the model does not become biased toward the majority class.

Optimizer

The AdamW optimizer is used for training the model. It is an extension of the Adam optimizer that includes weight decay, which helps improve generalization. A low learning rate is used during fine-tuning to preserve pretrained knowledge while adapting the model to the dataset.

Workflow

1. **Data:** The dataset used in this project consists of chest X-ray images categorized into Normal and Pneumonia classes. These images are collected and organized into training, validation, and test sets.
2. **Preprocessing:** All images are resized to 224×224 pixels to match the input size required by the models. Pixel values are normalized to the range [0,1] to improve training stability. Data augmentation techniques such as rotation, zooming, and horizontal flipping are applied to the training dataset to increase diversity and reduce overfitting.
3. **Training:** The models are trained on the training dataset with a validation split to monitor performance during training. Early stopping is used to prevent overfitting by stopping training when validation performance stops improving. Model checkpointing is also used to save the best-performing model.
4. **Testing:** After training, the models are evaluated on a separate test dataset that was not used during training. This ensures that the performance reflects the model's ability to generalize to unseen data.
5. **Evaluation:** The performance of the models is evaluated using multiple metrics including accuracy, confusion matrix, classification report (precision, recall, F1-score), and ROC-AUC score. These metrics provide a comprehensive understanding of the model’s performance.

Implementation Details

The implementation of the pneumonia detection system was carried out using deep learning frameworks in a Google Colab environment with GPU support. TensorFlow and Keras libraries were used to design, train, and evaluate the models. The dataset of chest X-ray images was loaded from Google Drive and organized into training, validation, and testing directories.

ImageDataGenerator was used for efficient data loading and preprocessing. It enabled real-time data augmentation and normalization of images. All images were resized to 224 × 224 pixels to match the input requirements of the deep learning models.

Four models were implemented in this study, including three base models—custom CNN, VGG16, and ResNet50—and one advanced model, DINOv2. The custom CNN was developed from scratch to serve as a baseline model. VGG16 and ResNet50 were implemented using transfer learning with pretrained weights from the ImageNet dataset. In VGG16, all convolutional layers were frozen to retain learned features, whereas in ResNet50, early layers were frozen and deeper layers were fine-tuned to adapt to the dataset.

In addition to the base models, an advanced DINOv2-based model was implemented. Unlike traditional convolutional architectures, DINOv2 is based on a Vision Transformer (ViT-B/14) architecture pretrained using self-supervised learning on a large-scale dataset. A two-stage fine-tuning strategy was employed, where initially only the classification head was trained (warm-up phase), followed by full model fine-tuning to improve performance on the pneumonia classification task.

Training was performed using mini-batch gradient descent with a batch size of 32. EarlyStopping and ModelCheckpoint callbacks were used to improve training efficiency and save the best-performing model. The models were evaluated using multiple metrics including accuracy, confusion matrix, classification report, and ROC-AUC score.

Hyperparameter Values

The following hyperparameters were used in this project:

General Hyperparameters

- Image Size: 224 × 224
- Batch Size: 32
- Number of Epochs: 10 (Base Models), 15 (+ 3 warm-up epochs for DINOv2)
- Optimizer: Adam (Base Models), AdamW (DINOv2)
- Loss Function: Binary Crossentropy (Base Models), Cross-Entropy with class weighting (DINOv2)
- Dropout Rate: 0.5 (Base Models), 0.3 (DINOv2)

Learning Rates

- Custom CNN: 0.001
- VGG16: 0.001
- ResNet50 (Fine-tuning): 0.0001
- DINOv2: Stage 1 (Head Warm-up): 0.001, Stage 2 (Full Fine-tuning): 0.00005

Data Augmentation

- Rotation Range: 10 degrees
- Zoom Range: 0.2
- Horizontal Flip: Enabled
- Colour Jitter (DINOv2): Brightness (0.2), Contrast (0.2)

Model-Specific Parameter Values

Custom CNN

- Filters: 32, 64, 128
- Dense Units: 128

VGG16

- Dense Units: 256
- Base layers frozen

ResNet50

- Dense Units: 256
- Global Average Pooling used
- Partial fine-tuning applied

DINOv2

- Backbone: Vision Transformer (ViT-B/14, 12 encoder blocks)
- Patch Size: 14 × 14
- Embedding Dimension: 768
- Classification Head Units: 256
- Activation Function: GELU
- Pretraining: Self-supervised on LVD-142M dataset
- Fine-tuning Strategy: Two-stage (head warm-up + full fine-tuning)
- Learning Rate Scheduler: Cosine Annealing
- Weight Decay: 0.0001

Results and Analysis

Evaluation Metrics

The performance of the implemented models was evaluated using standard classification metrics including accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of how well the models perform in detecting pneumonia from chest X-ray images.

Accuracy measures the overall correctness of the model, while precision indicates how many of the predicted positive cases were actually correct. Recall is particularly important in medical diagnosis as it reflects the ability of the model to correctly identify pneumonia cases. F1-score provides a balance between precision and recall.

Model	**Accuracy (in %)**	**Precision** (Normal)	**Recall** (Normal)	**Precision** (Pneumonia)	**Recall** (Pneumonia)	**F1-Score**
Custom CNN	87.98	0.94	0.72	0.85	0.97	0.88
VGG16	90.70	0.95	0.79	0.89	0.97	0.91
ResNet50	86.54	0.87	0.76	0.86	0.93	0.87
DINOv2	84.78	0.89	0.77	0.86	0.90	0.88

Table 1: Evaluation Metrics for Different Models

From the results, it can be observed that the VGG16 model achieved the highest accuracy and F1-score, indicating superior overall performance among all the evaluated models. The custom CNN also performed well and served as a strong baseline; however, it exhibited lower recall for normal cases, indicating a bias towards pneumonia detection.

ResNet50, despite being a deeper architecture, did not outperform VGG16. This can be attributed to the limited dataset size and the challenges associated with effective fine-tuning of deeper networks.

The DINOv2-based model, although a more recent and advanced architecture, achieved comparatively lower performance in terms of accuracy and AUC. While it demonstrated good recall for pneumonia cases, its overall performance was affected by instability during training and insufficient adaptation to the dataset. This suggests that such advanced models require larger datasets and more careful tuning to outperform traditional transfer learning approaches like VGG16.

Figures & Tables

Tables and graphical representations were used to compare the performance of different models effectively. The comparison table summarizes the key metrics such as ROC-AUC scores for all three models.

Model	AUC
Custom CNN	0.955
VGG16	0.964
ResNet50	0.943
DINOv2	0.917

Table 2: Area Under the Curve (AUC) for Different Models

The graphical analysis clearly shows that the VGG16 model provides the best balance between sensitivity and specificity among all the evaluated models. The ROC curves indicate that all models perform reasonably well in distinguishing between normal and pneumonia cases; however, VGG16 achieves the highest area under the curve (AUC), demonstrating superior classification capability.

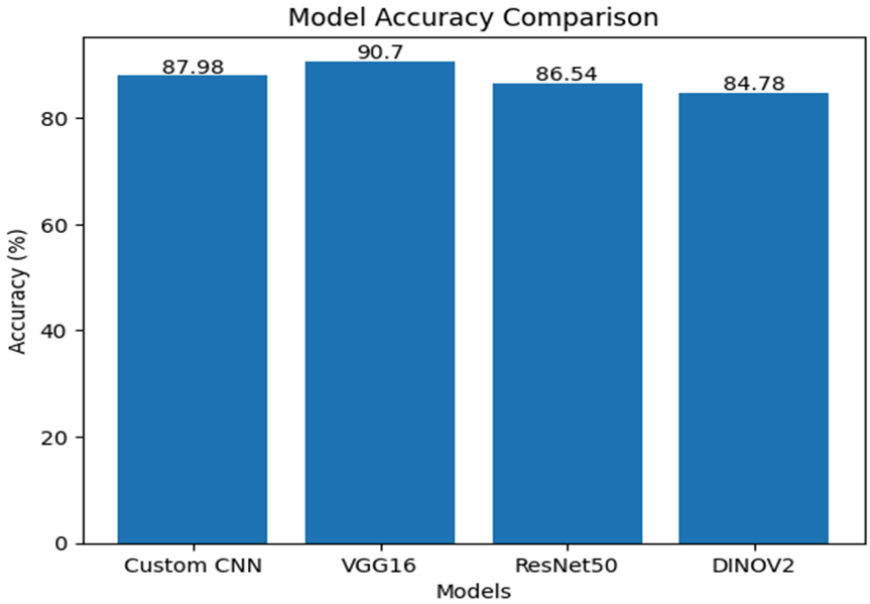

Fig. 6: Graph representing Accuracy for Different Models

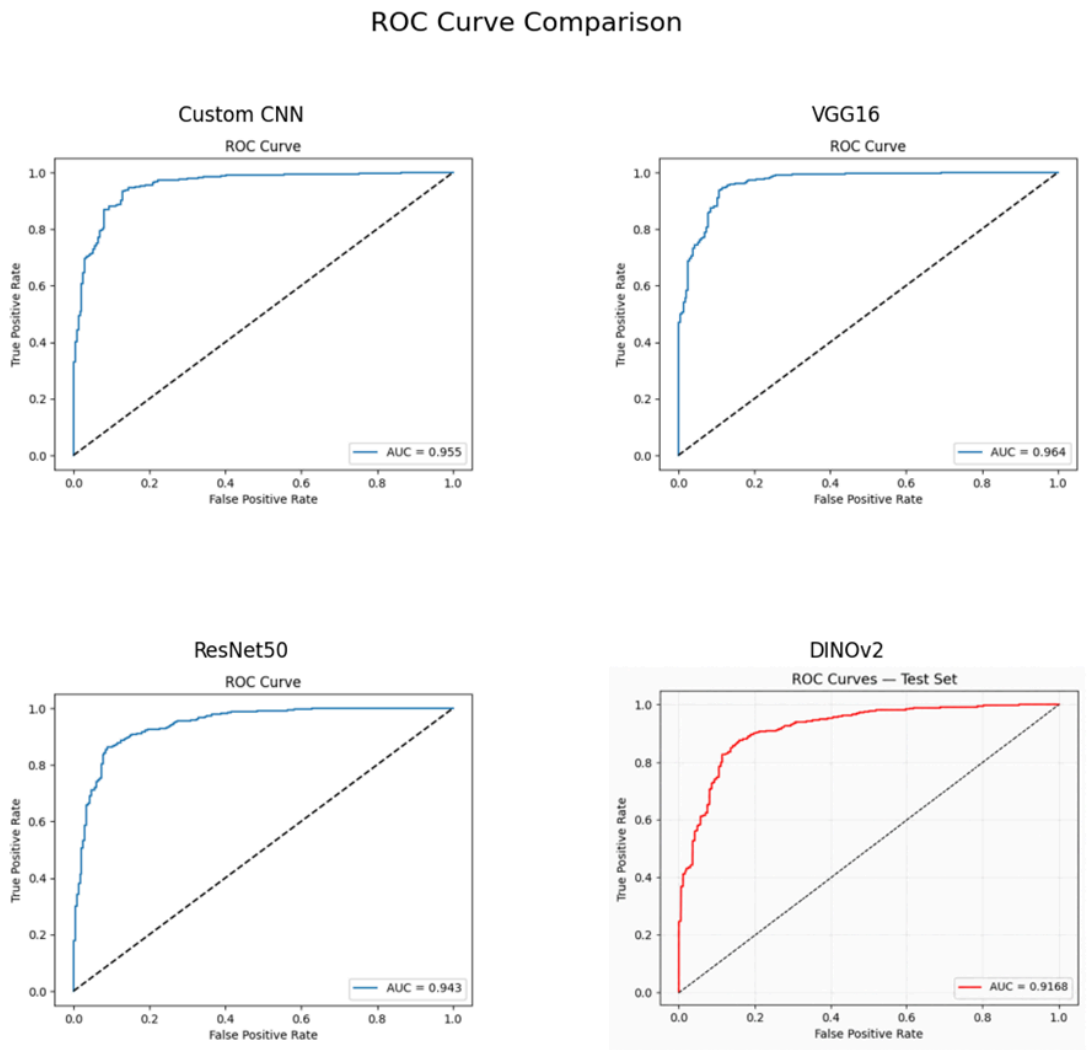

Fig. 7: Graphs showing ROC Curves for Different Models

The custom CNN and ResNet50 models also show strong performance, with high AUC values indicating good discriminative ability. In comparison, the DINOv2-based model achieves a relatively lower AUC score, suggesting comparatively weaker class separation. This further reinforces that, despite being a more advanced architecture, its performance is limited in this scenario due to dataset size and training constraints.

Loss Curves

Loss curves for both training and validation datasets were analyzed to understand the learning behavior of the models.

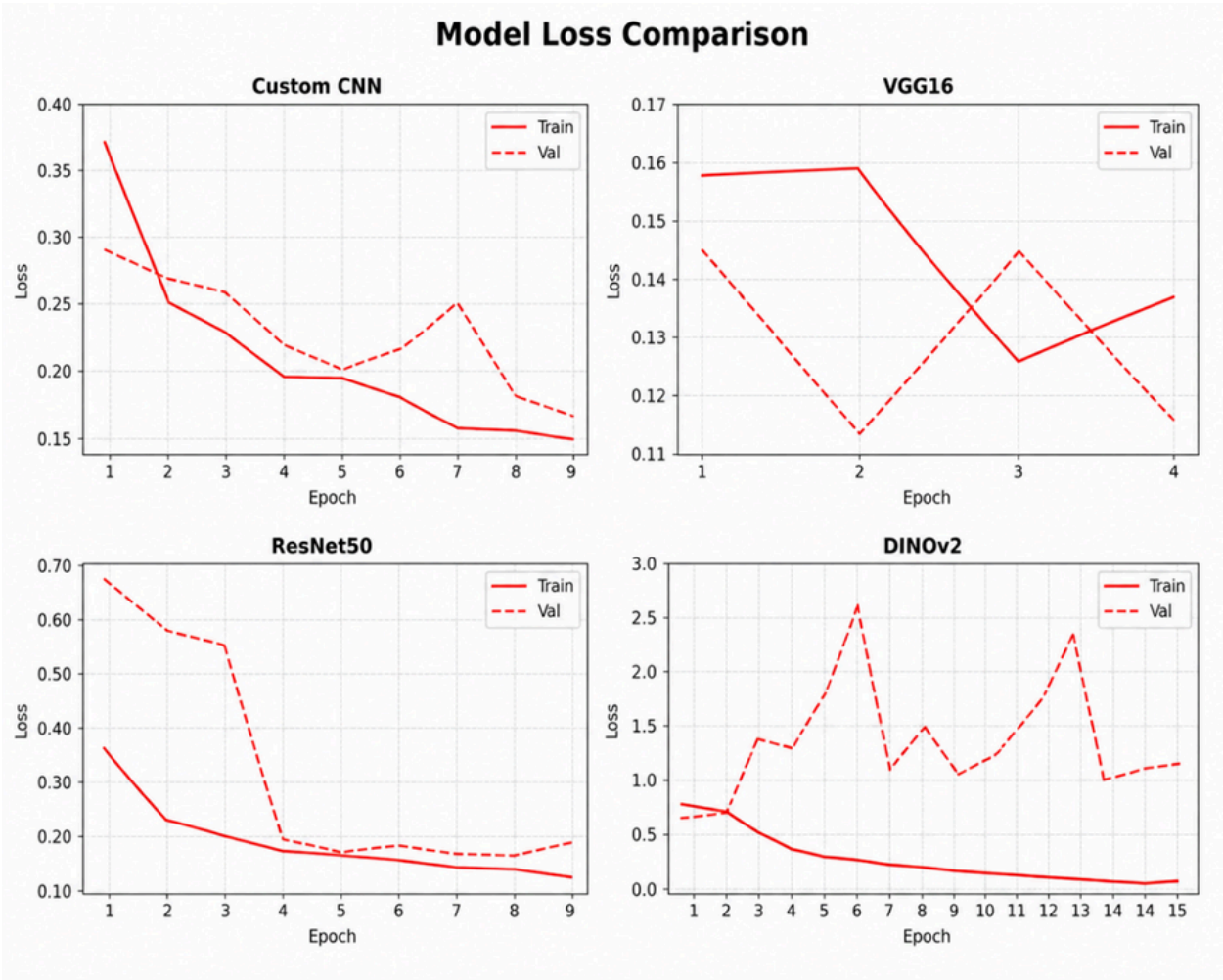

Fig. 8: Graphs showing Loss for Different Models

The loss curves indicate that the models generally converged during training, demonstrating effective learning of features from the dataset. In the case of the custom CNN and VGG16, the validation loss closely follows the training loss, suggesting minimal overfitting and good generalization performance.

ResNet50 shows noticeable fluctuations in validation loss, indicating some instability during the fine-tuning process, likely due to the complexity of the model and sensitivity to hyperparameter settings.

The DINOv2-based model exhibits even greater fluctuations in validation loss compared to the other models, reflecting instability during training and potential overfitting. This behavior suggests that the model requires more careful tuning and possibly a larger dataset to achieve stable convergence and improved performance.

Confusion Matrix

The confusion matrix provides a detailed breakdown of correct and incorrect predictions made by the models.

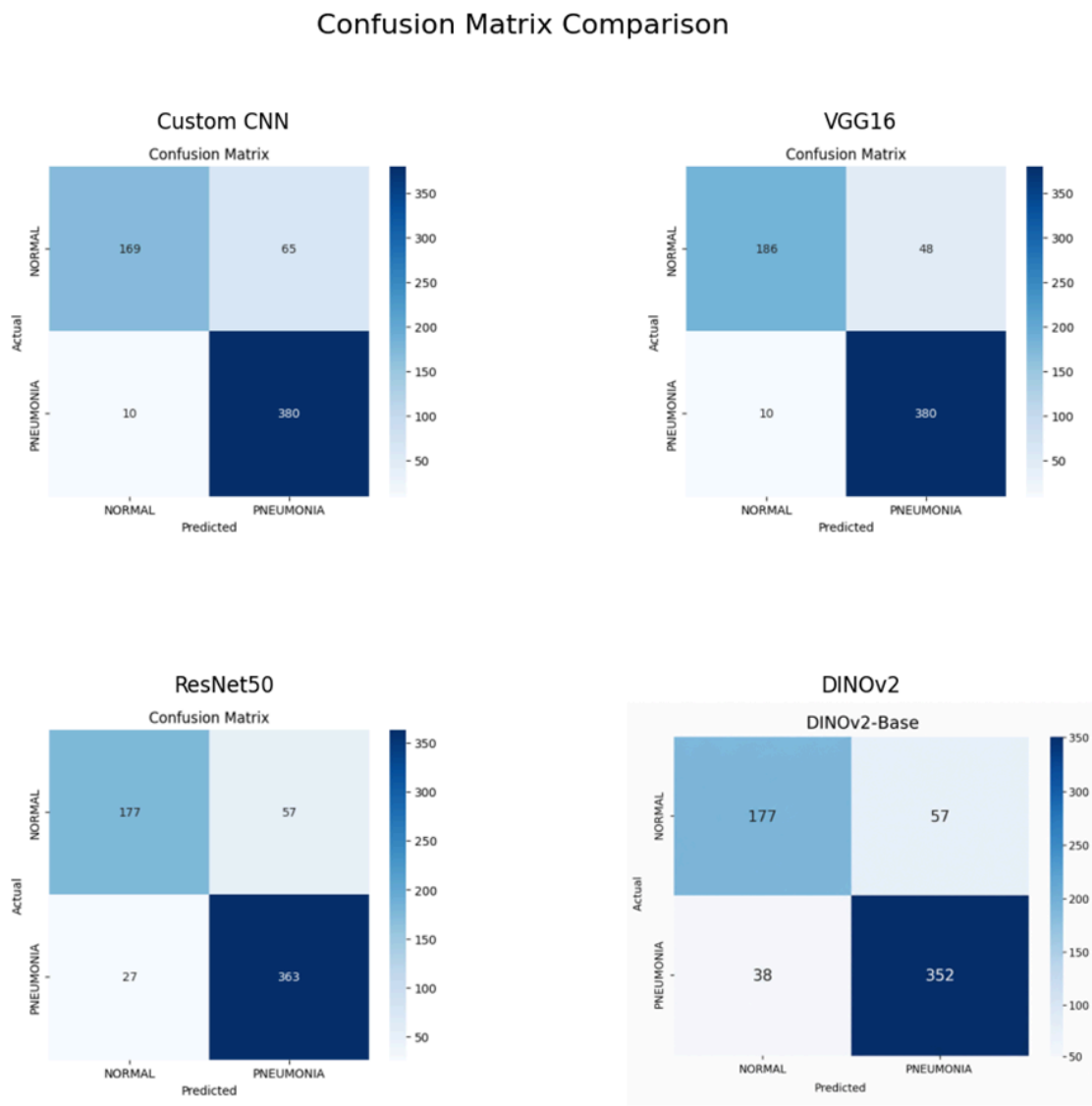

Fig. 9: Confusion Matrices for Different Models

From the confusion matrices, it is evident that all models perform well in detecting pneumonia cases, as indicated by their high true positive rates. This is particularly important in a medical context, where correctly identifying diseased cases is critical.

However, some misclassification is observed in identifying normal cases, especially in the custom CNN and ResNet50 models, where a higher number of normal images are incorrectly classified as pneumonia.

The VGG16 model demonstrates better balance, with fewer misclassifications and improved detection of both normal and pneumonia cases. In comparison, the DINOv2-based model, while maintaining good pneumonia detection, shows a higher number of false negatives and false positives, indicating comparatively weaker performance in accurately distinguishing between the two classes.

Comparison with Base Models

A comparative analysis was conducted between the base models, namely the custom CNN, VGG16, and ResNet50, and the advanced model, DINOv2.

The custom CNN serves as a baseline to evaluate the effectiveness of deeper architectures. Among the base models, VGG16 significantly outperformed both the custom CNN and ResNet50 due to its effective use of pretrained feature extraction, which enables better generalization on the dataset.

ResNet50, although deeper and more complex, did not achieve better results compared to VGG16. This can be attributed to the limited dataset size and the challenges associated with fine-tuning deeper networks.

The advanced DINOv2 model, despite being a more recent architecture, did not outperform the base models in this study. Its performance was affected by training instability and insufficient adaptation to the dataset, indicating that such advanced models require larger datasets and careful tuning to achieve optimal results.

Overall, the results demonstrate that among the base models, VGG16 is the most effective for this task, while the advanced model highlights the importance of dataset size and training stability in achieving superior performance.

Conclusions

This study demonstrates that deep learning models can significantly enhance pneumonia detection from chest X-rays. VGG16 achieved the highest diagnostic accuracy, while DINOv2-Base showcased the advantages of self-supervised learning and transformer-based architectures. Lightweight models such as CustomCNN remain valuable for deployment in resource-constrained settings.

Future work should focus on improving interpretability through explainable AI techniques and exploring hybrid architectures that combine CNN and transformer capabilities. The integration of such systems into clinical workflows has the potential to improve diagnostic accuracy and accessibility worldwide.

References

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